Answer Key - Physics - Undergraduate Level Assessment 13 (Answers are ordered exactly as in the question paper.)

Multiple Choice

1. **Answer:** D
   Options: A) (1/2) k T; B) kT; C) (3/2) k T; D) 3 kT
   Note: Correct option: D - 3 kT

2. **Answer:** C
   Options: A) 1.2 x 10^9 m/s; B) 3.0 x 10^8 m/s; C) 1.5 x 10^8 m/s; D) 1.0 x 10^8 m/s
   Note: Correct option: C - 1.5 x 10^8 m/s

3. **Answer:** D
   Options: A) exactly 10 times; B) an average of 10 times, with an rms deviation of about 0.1; C) an average of 10 times, with an rms deviation of about 1; D) an average of 10 times, with an rms deviation of about 3
   Note: Correct option: D - an average of 10 times, with an rms deviation of about 3

4. **Answer:** D
   Options: A) As; B) P; C) Sb; D) B
   Note: Correct option: D - B

5. **Answer:** B
   Options: A) Constant temperature; B) Constant volume; C) Constant pressure; D) Adiabatic
   Note: Correct option: B - Constant volume

True / False

6. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.

7. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.

8. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.

9. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.

10. **Answer:** False
    Options: A) True; B) False
    Note: LLM-generated false statement. Correct answer: '(3$^{(}$1/2))mc'.

Long Form Response

11. **Answer:** None. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
    Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 427 Hz. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
    Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key